

Area of trapezoid and polygon

trapeziumformula · Segmentation · Filling · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5, Part A, + Modern Education, Part 5, Unit 3-4

Core Trap: Correct or wrong size after segmentation

SSPA related:

 **High frequency** Presents the core of sub-test geometry questions, P6 basics of circular area Prerequisite knowledge:

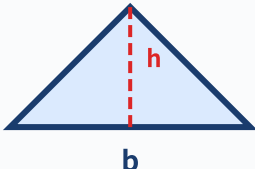
① Trapezoid area = (upper base + lower base) $\times$ height $\div$ 2 ② Find the area of polygons by division method ③

Find the area of polygons by filling method ④ Comprehensive solution to area word problems P4 area of square/rectangle · P5 area of parallelogram/triangle

Our goals: ① Trapezoid area = (upper base + lower base) $\times$ height $\div$ 2 ② Find the area of polygons by division method ③ Find the area of polygons by filling method ④ Comprehensive solution to area word problems

Student Name: _____ Class: _____ Date: _____ Time Spent: _____

I.Warm-Up Questions (total 5 question , 5 minutes) — reviewparallelogramandtrianglearea

#	Question	Difficulty	Working Space
1	The parallelogram has a base of 8 cm and a height of 5 cm. Area = ?		
2	Triangular base 10 cm, height 6 cm. Area = ?		
3	The rectangle is 12 cm long and 7 cm wide. Area = ?		
4	The area of a parallelogram is 48 cm ² and its base is 8 cm. High = ?		
5	The area of the triangle in the figure below = ? 		

II.Core Knowledge + Worked Examples

Knowledge point 1: Trapezoidal area formula SSPA required test

① **Trapezoid** |||SEP|||: A quadrilateral with only one pair of parallel sides. The two parallel sides are called upper base and lower base
Area formula: trapezoid area = (upper base + lower base) × height ÷ 2

② ③ **Height** =

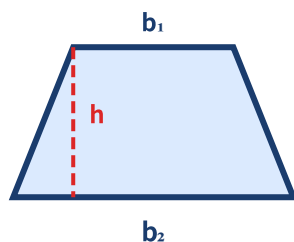
vertical distance between upper base and lower base(not the hypotenuse!)④ Two identical trapezoids can be combined into a parallelogram → so the formula is

⑤ Reverse: Height = Area × 2 ÷ (Upper Base + Lower Base); Upper Base + Lower Base = Area × 2 ÷ Height ÷ 2

⑤ Reverse calculation: height = area × 2 ÷ (upper base + lower base); upper base + lower base = area × 2 ÷ height

Trapezoidal area = (upper base + lower base) × height ÷ 2

Two identical trapezoids → parallelogram



Two identical trapezoids
form a parallelogram

the area of the trapezoid must be divided by

Most frequent error: forgetting ÷ 2! The formula for a trapezoid is different from that of a parallelogram and must be divided by 2.

(Upper base + lower base) ÷ 2 × height has the same result as (upper base + lower base) × height ÷ 2, but it is recommended to multiply the height first and then ÷ 2 to avoid errors in fraction calculations.

Example 1

The upper base of the trapezoid is 5 cm, the lower base is 9 cm, and the height is 4 cm. Area = ?

(substitute directly: $(5+9) \times 4 \div 2$)

Example 2

The area of the trapezoid is 42 cm^2 , upper base + lower base = 14 cm. High = ?

(reverse calculation: height = $42 \times 2 \div 14 = 6 \text{ cm}$)

— PAGE 3: KP1 continuation + KP2 —

Example 3

The upper base of the trapezoid is 7 cm, the lower base is 13 cm, and the height is 6 cm. Area = ?

(Note: The question may also give the length of the hypotenuse 10 cm - that is redundant information! Only take the top, bottom, bottom and height)

Example 4

The trapezoid has an area of 80 cm^2 , a height of 8 cm, and a top and bottom of 6 cm. Bottom = ?

(First find upper bottom + lower bottom = $80 \times 2 \div 8 = 20$, lower bottom = $20 - 6 = 14 \text{ cm}$)

Knowledge point 2: Polygon area - segmentation method ● SSPA

① Segmentation method |||SEP|||: Cut the irregular polygon into several simple shapes (rectangle, square, triangle, trapezoid) ② Calculate the area of each simple shape separately, and then

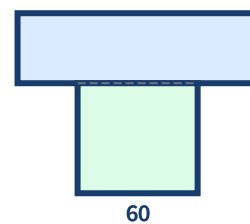
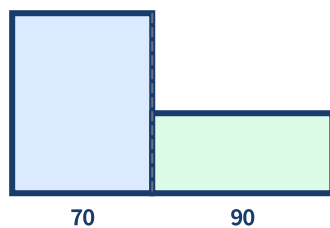
add them all up ③ Key: The dividing line should be drawn on

Know the position of the size to ensure that each small shape can be calculated ④ Tip: "It can be cut horizontally or vertically. The dimensions must be aligned after cutting. Mark A B C first, count each one and then add them up."

④ Tip: "You can cut horizontally or vertically. The dimensions must be aligned after cutting. Mark A B C first, count them one by one, and then add them up."

L-shaped dividing method: cut into two rectangles

Convex-shaped segmentation: cut into three pieces



Example 5

L-shaped pattern (refer to the left picture above): the vertical part is 12 cm long and 6 cm wide, and the horizontal part is 8 cm long and 4 cm wide. Find the total area using the division method.

Example 6

Convex shape (refer to the right picture above): the upper raised part is $3\text{ cm} \times 2.5\text{ cm}$, the middle horizontal bar is $12\text{ cm} \times 3.2\text{ cm}$, and the lower body is $8\text{ cm} \times 5\text{ cm}$. Total area = ?

Knowledge point 3: Polygon area - filling method 📌 SSPA

- ① **Filling method** |||SEP|||: Fill the irregular shape into a large simple shape, and then subtract the excess part
② Suitable for graphics with "gaps" and "concaves"
③ The formula: "First make up to make it generous, and then reduce the excess. Make up to make it complete, and reduce to make it accurate."
② Suitable for graphics with "gaps" and "concaves"
③ Tip: "First make up to make it generous, and then reduce the excess. Make up to make it complete, and reduce to make it accurate."

Instructions for filling method: concave shape



Use "Add" for the division method and "Subtract" for the filling method. Look at the graph to decide which one is more convenient! It is faster to use the filling method for recessed areas.

Example 7

Concave-shaped pattern (refer to the picture above): The outer frame is a large rectangle of $14\text{ cm} \times 8\text{ cm}$, and a small rectangle of $6\text{ cm} \times 4\text{ cm}$ is cut out in the middle. Find the remaining area.

Knowledge point 4: Area word questions 📌 SSPA must take the exam

- ① **Determine the shape first** (Trapezoid? L-shape? Combined shape?) → Select the corresponding method (formula/split/fill)
② **The units must be consistent** |||SEP|||: Convert different units first ($\text{cm} \rightarrow \text{m}$, $\text{mm} \rightarrow \text{cm}$)
③ At the end of the application question, you must write Answer and unit unit

Example 8

The upper base of the trapezoidal garden bed is 6 m , the lower base is 10 m , and the height is 5 m . If 3 flowers are planted per m^2 , how many flowers can be planted in total?

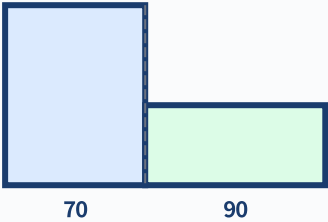
(First find the area → then multiply by the number of trees per m^2 → write the answer)

III. Lesson Layered Synchronization Practice

Basic layer (required by everyone, total 5 questions)

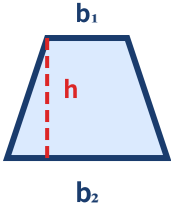
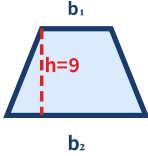
#	Question	Difficulty	Working Space
6	The upper base of the trapezoid is 6 cm, the lower base is 10 cm, and the height is 4 cm. Area = ?		
7	The upper base of the trapezoid is 5 cm, the lower base is 8 cm, and the height is 6 cm. Area = ?		
8	The upper base of the trapezoid is 9 cm, the lower base is 11 cm, and the height is 5 cm. Area = ?		
9	The L-shaped pattern can be divided into two rectangles: A (8×3 cm) and B (5×4 cm), with total area = ? (Segmentation method: first calculate the area of A, then calculate the area of B, and add them together)		
10	Take a large rectangle of $15 \text{ cm} \times 10 \text{ cm}$ and cut out a small rectangle of $7 \text{ cm} \times 4 \text{ cm}$. Remaining area = ? (filling method: large – small square)		

Advanced layer (choose do, total 5 questions)

#	Question	Difficulty	Working Space
11	The trapezoid has an area of 56 cm^2 , a height of 7 cm, and a top and bottom of 6 cm. Bottom = ?		
12	Trapezoidal upper base + lower base = 18 cm, height = 5 cm. Area = ? (You don't need to know the upper and lower bottoms separately! Use and substitute directly)		
13	The area of the combined shape (left rectangle + right rectangle) in the figure below = ? 		
14	The picture below shows a triangle added to a rectangle. The total area = ? 		
15	The trapezoid has an area of 96 cm^2 , a lower base of 14 cm, and an upper base of 10 cm. High = ?		

challenge layer (choose do, total 5 questions)

#	Question	Difficulty	Working Space
16	The upper base of a trapezoid is half the lower base. Bottom 12 cm, height 5 cm. Area = ? (Find the upper and lower parts first = $12 \div 2 = 6 \rightarrow$ Substitute into the formula)		

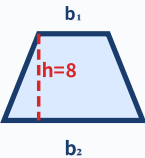
17	The upper base of the isosceles trapezoid in the picture on the right is 4 cm, the lower base is 8 cm, and the height is 6 cm. Area = ? 		
18	Combined graphics: A large rectangle of 15×10 cm can be made up, minus a 5×4 cm gap. Remaining area = ?		
19	 The trapezoid has an area of 108 cm^2, a height of 9 cm, and the upper base is 4 cm less than the lower base. How much are the upper and lower bases? (First find upper base + lower base = $108 \times 2 \div 9 = 24$, set lower base = x upper base = $x - 4 \rightarrow 2x - 4 = 24$)		
20	 Concave glyph area = ? The outer frame is $20 \text{ cm} \times 12 \text{ cm}$, and the middle rectangular gap is $10 \text{ cm} \times 6 \text{ cm}$.		

IV.areaapplicationquestion (total 6 question)

#	Question	Difficulty	Working Space
21	The upper base of the trapezoidal garden bed is 4 m, the lower base is 8 m, and the height is 5 m. Garden area = ? m^2		
22	A trapezoidal wall has an upper base of 3 m, a lower base of 7 m, and a height of 4 m. To paint, use 0.2 L of paint per m^2 . How many liters are needed in total?		
23	An L-shaped living room that can be divided into two rectangles: $6 \text{ m} \times 4 \text{ m}$ and $3 \text{ m} \times 5 \text{ m}$. Total living room area = ? m^2		
24	The terraced field is 20 m above ground, 30 m below ground, and 15 m high. Four vegetables can be planted per m^2 . How many vegetables can be planted in total?		

25	 A polygonal square = trapezoid + rectangle. Trapezoid: upper base 5 m, lower base 9 m, height 4 m. Rectangle: 8 m × 6 m. Total area = ?		
26	An irregular piece of land, with a 4 m × 3 m parking space dug out from a 15 m × 10 m rectangle. Remaining area = ? m ²		

V. Ultimate challenge area (total 3 questions, choose do, SSPA finale + competition level)

#	Question	Difficulty	Working Space
 1	 Trapezoidal area 120 cm², height 8 cm. The upper sole is 6 cm less than the lower sole. How much are the upper and lower bases? (Set lower base=x, upper base=x-6 → (2x-6) × 8 ÷ 2 = 120 → solve x)		
 2	A pentagon can be divided into three parts: triangle (base 10, height 4) + rectangle (10 × 6) + trapezoid (base 6, base 10, height 3). Total area = ? Calculate into three parts: Triangle A = Base x Height/2 Rectangle B = Length 梯形 C = (a+b) × h/2 總面積 = A + B + C		
 3	A trapezoid has the upper base twice as high as the lower base and the height as three times the lower base. Bottom 4 cm. Area = ? (Calculation: upper bottom=8, height=12 → (8+4) × 12 ÷ 2)		

VI. Class after homework

Basic must-do (total 5 questions)

#	Question	Difficulty	Working Space
H1	The upper base of the trapezoid is 7 cm, the lower base is 13 cm, and the height is 5 cm. Area = ?		
H2	The upper base of the trapezoid is 4 cm, the lower base is 10 cm, and the height is 8 cm. Area = ?		

H3	The upper base of the trapezoid is 8 cm, the lower base is 12 cm, and the height is 5 cm. Area = ?		
H4	The L shape can be divided into two rectangles of 9×3 cm and 6×5 cm. Total area = ?		
H5	Large rectangle 14×10 cm, cut out 6×4 cm. Remaining area = ?		

For advanced, choose do (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	The area of the trapezoid is 63 cm^2 , upper base + lower base = 18 cm. High = ?		
H7	The upper base of the trapezoid is 3 times the lower base, which is 5 cm and the height is 6 cm. Area = ?		
H8	The trapezoid has an area of 90 cm^2 , a height of 6 cm, and the lower base is twice as large as the upper base. How much are the upper and lower bases?		

VII. The Lessoncorecommon errorssummary

 **Review of Learning Objectives - After completing this lesson you should be able to:**

- ☐ Identify all trap types in our hall
- ☐ Solve  basic questions independently (100% correct)
- ☐ Challenge  Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Forget the trapezoid $\div 2$ SEP : Treat it as a parallelogram: Treated as a parallelogram	Trapezoid = (upper base + lower base) $\times$ height SEP . Must $\div 2$! $\div 2$. Must $\div 2$!
2	(upper bottom + lower bottom) $\div 2$ and forget to multiply by height	Three steps of the complete formula: (upper bottom + lower bottom) $\rightarrow \times$ height $\rightarrow \div 2$, no steps can be missed
3	High use of wrong hypotenuse length	Height= vertical distance between two bases, not the hypotenuse!
4	The size after division is correct SEP : The position of the dividing line is incorrect, resulting in the wrong size: Incorrect position of dividing line leads to wrong size	After division, the side length of each small shape must be inferred from the original drawing dimension.
5	Filling method to reduce errors SEP : error in generous size or gap size: Wrong square size or notch size	Confirm whether the "generous" and "part to be reduced" dimensions correspond to the original image
6	Superfluous information interference (T4 trap) SEP : Unnecessary data is used: Used unnecessary data	Determine the shape $\rightarrow$ Determine what data is needed $\rightarrow$ Use only necessary data and ignore interference items
7	A certain part of the combined figure is omitted	Mark A, B, C one by one, making sure each block is calculated and added up